

Multimedia Computing

By

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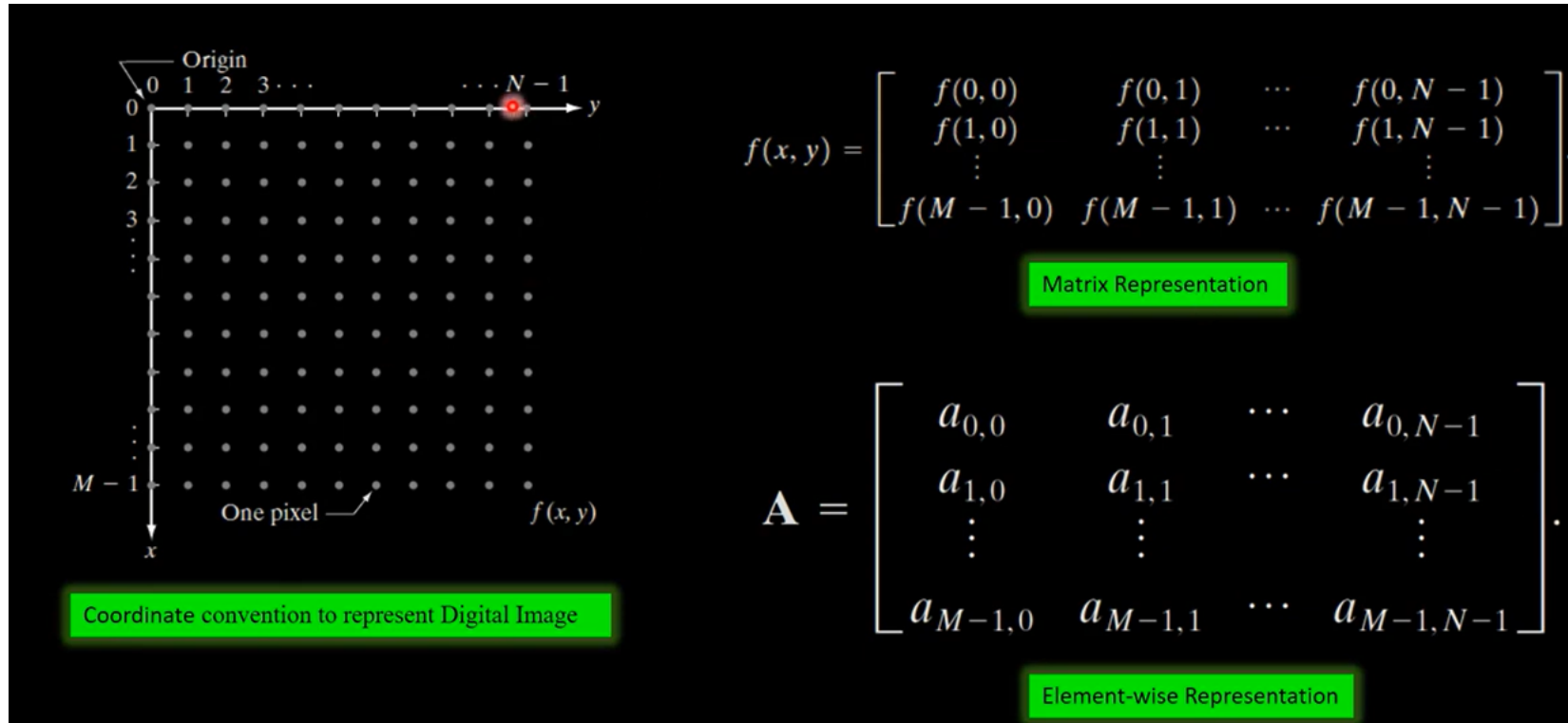
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Content

- ▶ Picture element
- ▶ Basic relationship among pixels
- ▶ Neighbors of a pixel
- ▶ Connectivity
- ▶ Adjacent pixel
- ▶ Arithmetic operation
- ▶ Color representation
- ▶ CIE XYZ to RGB

Coordinate Convention



Matrix Flattening

- ▶ For performance reasons we usually do not store image in the matrix fashion when coding image processing algorithms in Embedded Systems.
- ▶ There are two prevalent ordering schemes for storing flattened 2D matrices:
 - ❖ Row-major
 - ❖ Column-major

row-major	column-major
$f[0][0]$	$f[0][0]$
$f[0][1]$	$f[1][0]$
$f[0][2]$	$f[2][0]$
...	...
$f[0][N-1]$	$f[M-1][0]$
$f[1][0]$	$f[0][1]$
$f[1][1]$	$f[1][1]$
$f[1][2]$	$f[2][1]$
...	...
$f[1][N-1]$	$f[M-1][1]$
...	...
$f[M-1][N-1]$	$f[M-1][N-1]$

Radiometry

- ▶ In radiometry light is considered to be a traveling form of energy and therefore the unit used to describe, the unit of energy, *joule (J)*. This energy is associated with a source of origin, which is usually defined by its position, a direction of propagation and a wavelength λ . The particle theory of light where the smallest unit of light is considered to be a photon or a quantum of energy. This also conforms to the wave theory of light, which assumes light to be a waveform traveling in a particular direction. λ is expressed by units of nanometers (*nm*). Light travels with a speed c (*vacum*) n in a medium of refractive index n . An invariant in this context is the frequency f of light is given by

$$f = \frac{c_n}{\lambda},$$

Radiometry

- ▶ which does not change unlike C_n and λ . Another invariant is the energy carried by a photon

$$q = \frac{hc}{\lambda}$$

The *spectral energy* is a density function that gives the density of the quantum energy at an infinitesimal interval of wavelengths around λ with a width of $\Delta\lambda$.

Radiometry is the science and technology of the measurement of radiation from all wavelengths within the optical spectrum. The basic unit of power in radiometry is the watt (W).

Photometry

- Study of measurement of light in terms of visual response it produces
- Study concerned with the ways of measuring intensity of light
- Branch of Radiometry

The difference between radiometry and photometry is that radiometry includes the entire optical radiation spectrum (and often involves spectrally resolved measurements), while photometry deals with the visible spectrum weighted by the response of the eye.

Visual sensitivity

- Sensitivity of an individual to particular wavelength at a given viewing condition
- Different for different individuals
- Variation can be achieved from the sensitivity /luminosity curve

Scotopic and photopic vision

➤ Photopic vision

- ❖ Maximum visual effect at 555 nm (yellow-green)
- ❖ At 510 nm and 610 nm the sensitivity reduces to half

➤ Scotopic Vision

- ❖ Maximum effect at 507 nm (blue-green)
- ❖ Lower intensities are needed

Photometry and Color

- ▶ Photometric quantities have an aspect of perception associated with them. Color is a photometric quantity that we use all the time. A person, who lost color perception due to an accident exclaimed with anguish, "My dog looks gray, tomato juice is black and color TV is a hodge podge". Color not only adds beauty to our life, but serves important signaling functions. The natural world provides us with many signals to identify and classify objects. Many of these come in terms of color. For e.g. banana turns yellow when it is ripe, the sky turns red when it is dawn and so on.

Basic information on Light

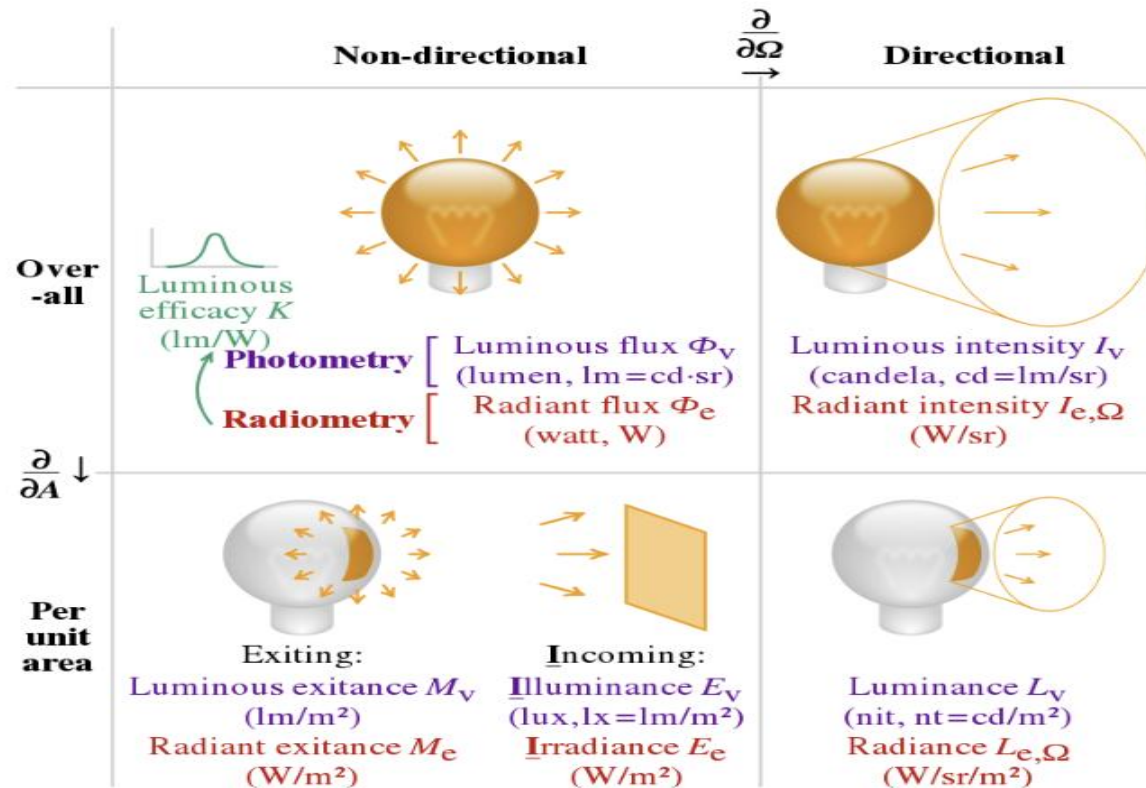
- ▶ The color of an object are determined by the nature of the light reflected
- ❖ A body that reflects light relatively balanced in all visible wavelengths appears **white** to the observer
- ❖ Whichever favors reflectance in a limited range of the visible spectrum exhibits some shades of color (chromatic light)

Achromatic light

- ▶ Light that is **void of color** is called **monochromatic (achromatic)** light.
- ▶ The only attribute of monochromatic light is its **intensity** or amount.
- ▶ Achromatic images known as **gray scale image**.

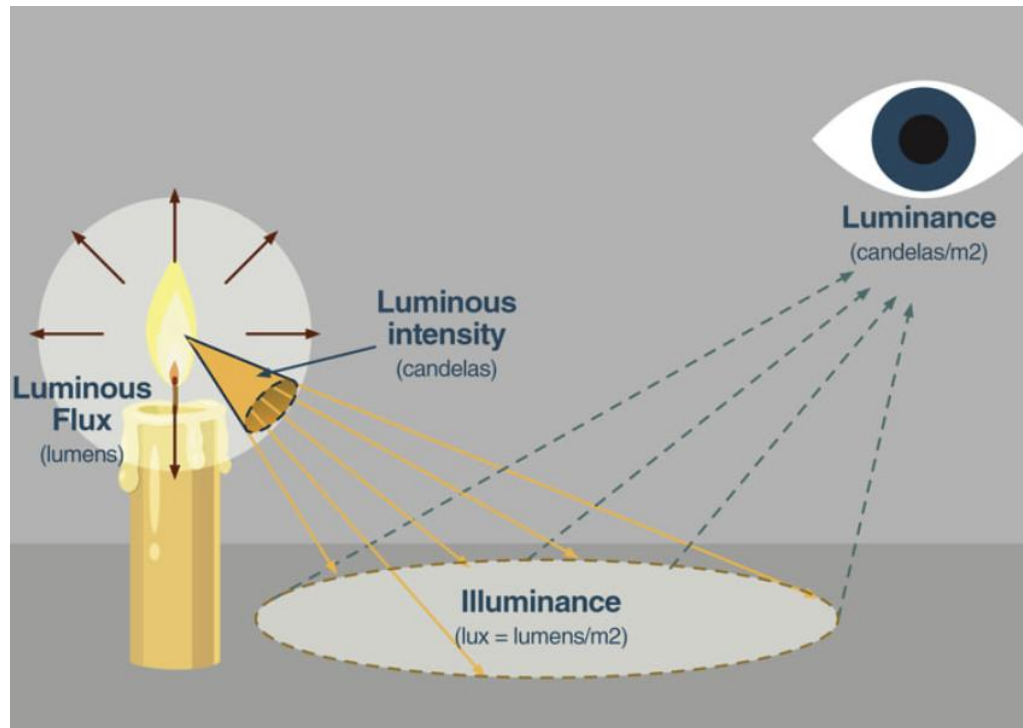
Radiance

- **Radiance** is the **total amount of energy** that flows **from the light source** , and it is usually measured in watts(W).



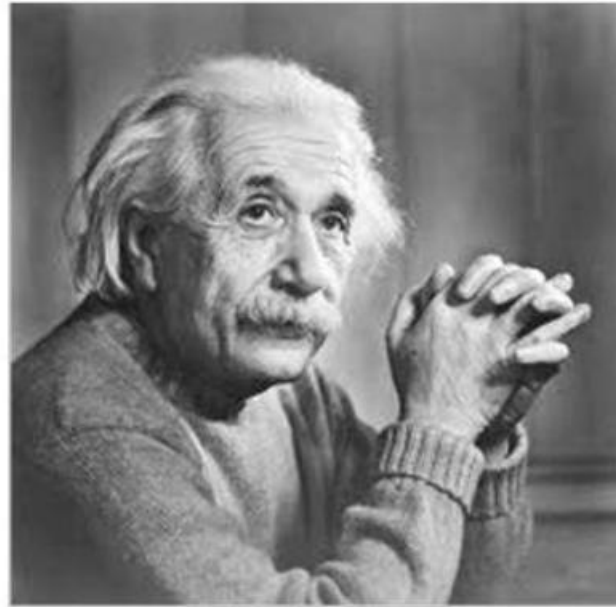
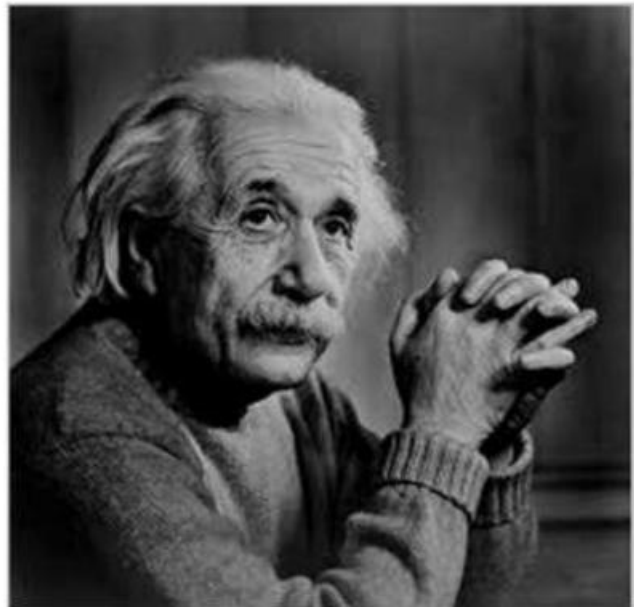
Luminance

- ▶ **Luminance**, measured in **lumens (lm)**, gives a measure of the **amount of energy an observer perceives** from a light source.



Brightness

- ▶ Brightness is a **subjective** descriptor of light perception that is practically **impossible to measure**.



Energy Source of Image

- ▶ The principal energy source for images in use today is the electromagnetic (EM) energy spectrum.
- ▶ Synthetic images, used for modeling and visualization are generated by computer.



Two Major Tasks

- ▶ Improvement of pictorial information for human interpretation
- ▶ Processing of image data for storage, transmission and representation for autonomous machine perception.

Image formation model

- ▶ Image refers to a 2D light-intensity function, $f(x,y)$
- ▶ Amplitude of f at spatial coordinates (x,y) gives the intensity (brightness) of the image at that point.
- ▶ Light is a form of energy thus $f(x,y)$ must be nonzero and finite.

$$0 < f(x,y) < \infty$$

Illumination and Reflectance

- ▶ The basic nature of $f(x,y)$ may be characterized by 2 components:
- ▶ Illumination, $i(x,y)$: the amount of source light incident on the scene being viewed.
- ▶ Reflectance, $r(x,y)$: the amount of light reflected by the objects in the scene.

Resolution

- ▶ Resolution (how much you can see the detail of the image) depends on sampling and gray levels.
- ▶ The bigger the sampling rate ($M \times N$) and the gray scale level (K), the better the approximation of the digitized image from the original. But the size of the image gets bigger

$$B = M \times N \times K$$

Due to storage and quantizing hardware consideration, the number of intensity level (L) typically is an integer power of 2.

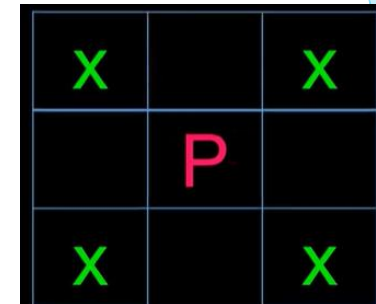
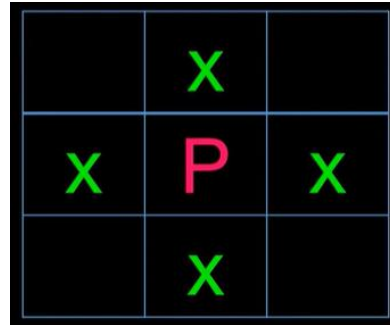
$$L = 2^k$$

Neighbor of a pixel

- ▶ A pixel P has at coordinates (x,y) has
- ▶ N4(p): four neighbors of p
- ▶ ND(P): four diagonal neighbor of P
- ▶ N8(p): Eight neighbors of P

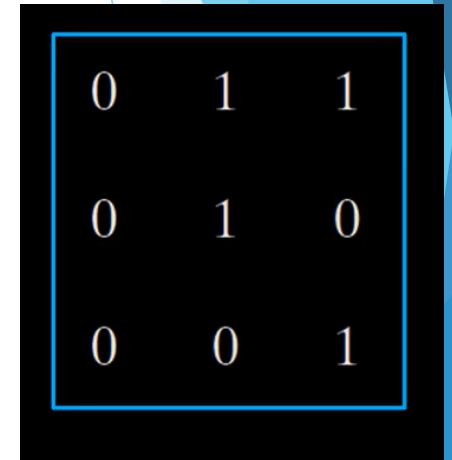
$(x+1, y), (x-1, y), (x, y+1), (x, y-1)$

$(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)$



Connectivity

- ▶ Let V be the set of gray-level values used to defined connectivity
- ▶ 4-connectivity:
2 pixels p and q with values from V are 4-connected if q is in the set $N_4(p)$
- ▶ 8-connectivity:
2 pixels p and q with values from V are 8-connected if q is in the set $N_8(p)$



0	1	1
0	1	0
0	0	1

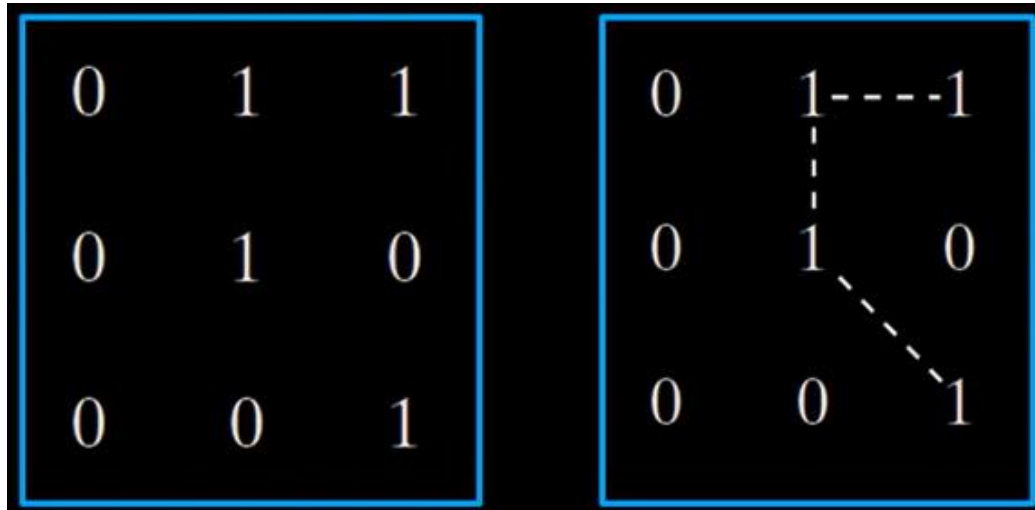
Connectivity

- ▶ Let V be the set of gray-level values used to define connectivity.
- ▶ M-connectivity(mixed connectivity)
- ▶ 2 pixels p and q with values from V are m connectivity if
 - ❖ q is the set $N_4(p)$ or

✓ q is in the set $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ is empty.
✓ (the set of pixels that are 4-neighbors of both p and q whose values are from V)

M-Connectivity

- ▶ m-connectivity eliminates the multiple path connections that arises in 8-connectivity.



Path

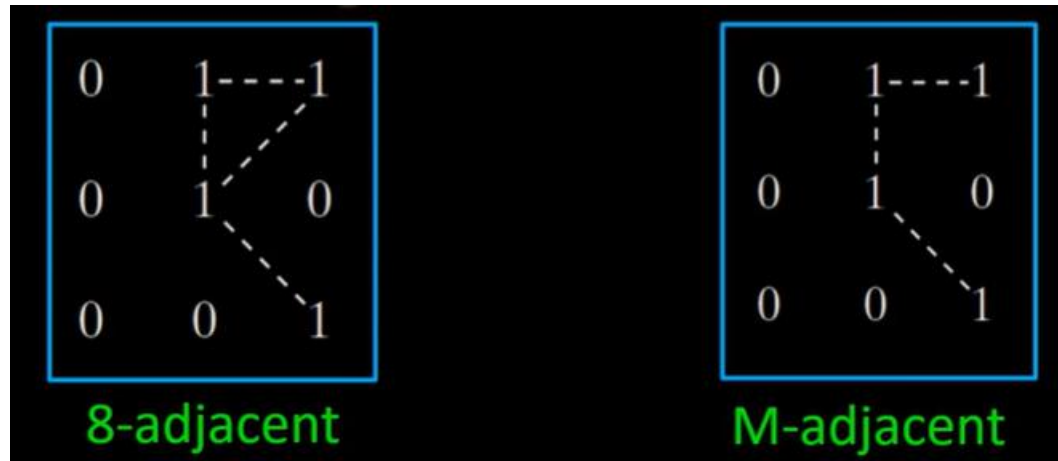
- ▶ A (digital) path from pixel P with coordinates (x,y) to pixel q with coordinates (s,t) is a sequence of distinct pixels with coordinates

$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$
where $(x_0, y_0) = (x, y)$, $(x_n, y_n) = (s, t)$ and (x_i, y_i) is adjacent to (x_{i-1}, y_{i-1})

- ▶ n is the length of the path

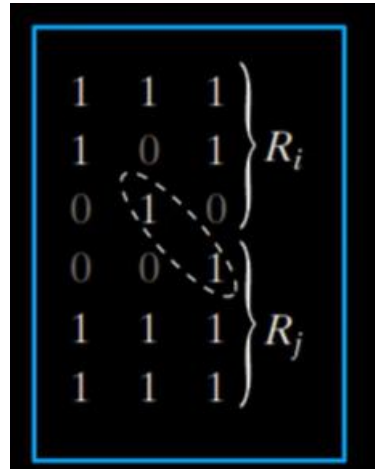
Adjacent

- ▶ A pixel p is adjacent to a pixel q if they are connected.



Adjacent

- ▶ Two image area subsets R_1 and R_2 are adjacent if some pixels in R_1 is adjacent to some pixels R_2



Adjacent

- ▶ The circle point is part of the boundary of the 1 valued pixels only if 8-adjacency between the region and background is used.



Distance measures

- ▶ For pixel p, q, z with coordinates $(x, y), (s, t)$ and (u, v) respectively,
- ▶ D is the distance function or metric if

(a) $D(p, q) \geq 0$; $D(p, q) = 0$ iff $p = q$
(b) $D(p, q) = D(q, p)$
(c) $D(p, z) \leq D(p, q) + D(q, z)$

Euclidean Distance

- ▶ The Euclidean distance is defined as the distance between two points. In other words, the Euclidean distance between two points in the Euclidean space is defined as the length of the line segment between two points.

Thus, the Euclidean distance formula is given by:

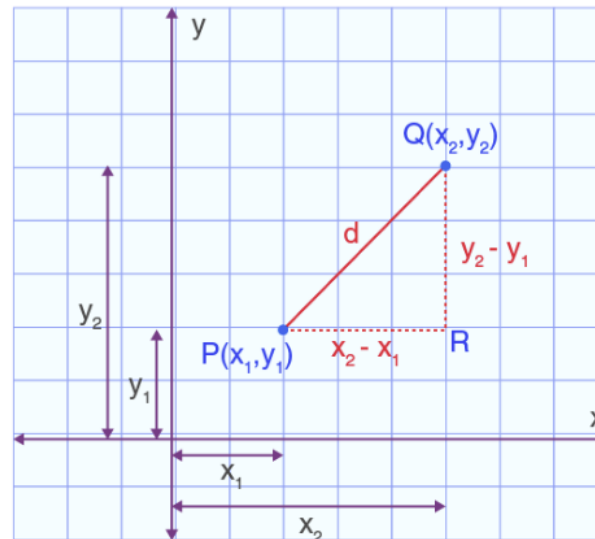
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where,

"d" is the Euclidean distance

(x_1, y_1) is the coordinate of the first point

(x_2, y_2) is the coordinate of the second point.



City block distance

- ▶ The D_4 distance (also called city-block distance) between P and q is defined as

$$D_4(p, q) = |x - s| + |y - t|$$

- ▶ In this case, the pixel having a distance from (x, y) less than or equal to some value r from a diamond centered at (x, y)

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

The pixels with $D_4=1$ are the 4-neighbors of (x, y) .

Chessboard Distance

- ▶ The D8 distance (also called chessboard distance) between p and q is defined as

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

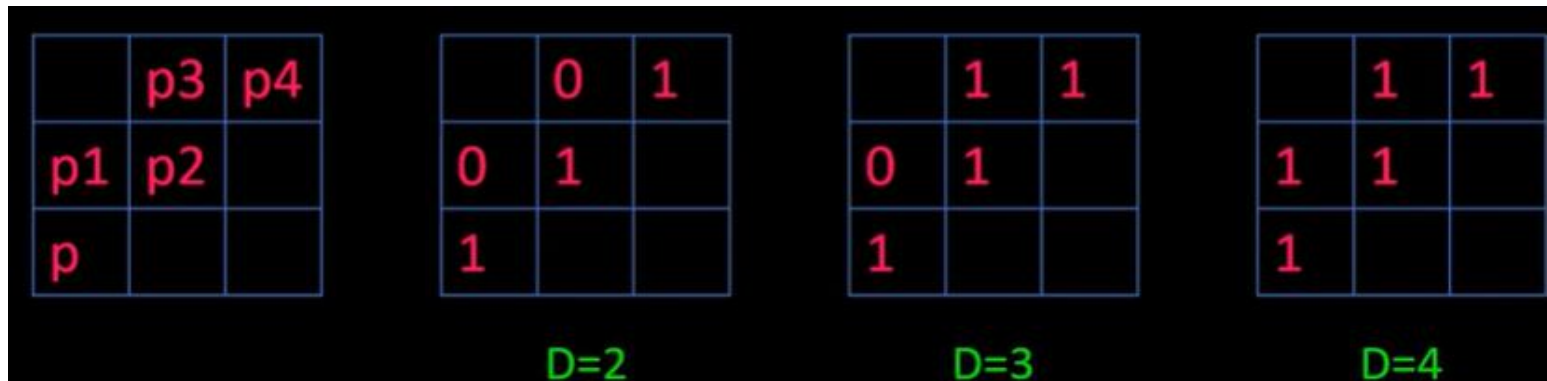
- ▶ In this case, the pixels with D8 distance from (x,y) less than or equal to some value r form a square centered at (x,y).

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

The pixels with $D_8=1$ are the 8-neighbors of (x, y).

M-connectivity distance

- ▶ Distance of m-connectivity of the path between 2 pixels depends on values of the pixels along the path.
- ▶ E.g. if only connectivity of pixel values 1 is allowed. Find the m-distance between p and p4



Arithmetic operation

Image arithmetic involves images are same size

$$s(x, y) = f(x, y) + g(x, y)$$

$$d(x, y) = f(x, y) - g(x, y)$$

$$p(x, y) = f(x, y) \times g(x, y)$$

$$v(x, y) = f(x, y) \div g(x, y)$$

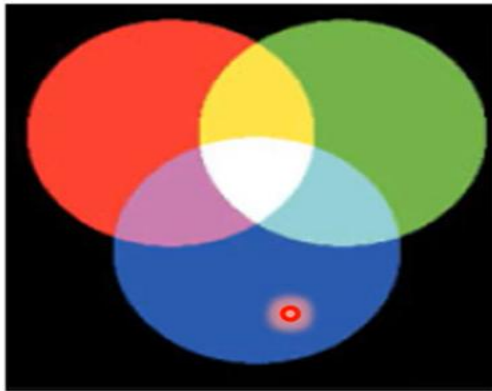
Logical operations

- ▶ Illustrations of logical operations involving foreground white pixels
- ▶ Black represent binary 0 and white represent binary 1



Color Image

- ▶ Separate channel for different color
 - ❖ RGB
- ▶ Some model separates color information from Brightness intensity
 - ❖ HSI



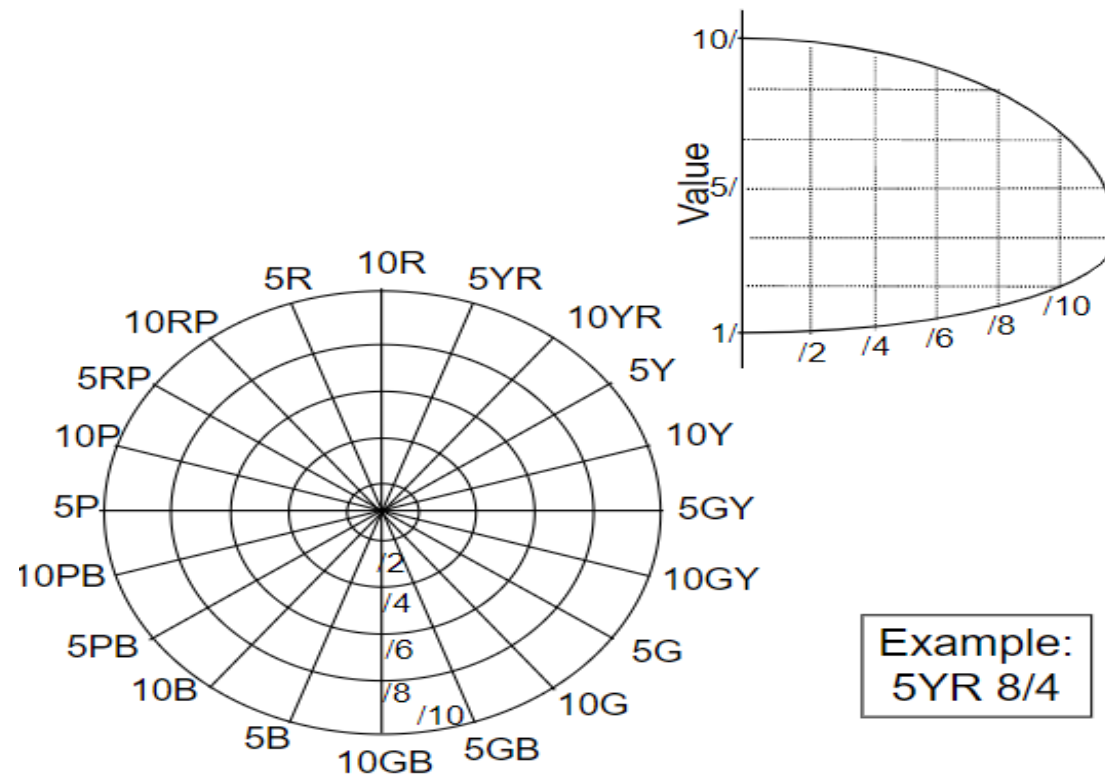
Munsell Color System (1915)

Equal perceptual steps in Hue Saturation Value.

Hue: R, YR, Y, GY, G, BG, B, PB, P, RP
(each subdivided into 10)

Chroma: 0 ... 20 (neutral ... saturated)

Value: 0 ... 10 (dark ... pure white)



Three dimensions of color

- ▶ Our perception of color can be described through 3 qualities:
 - HUE
 - Value
 - Intensity / Saturation

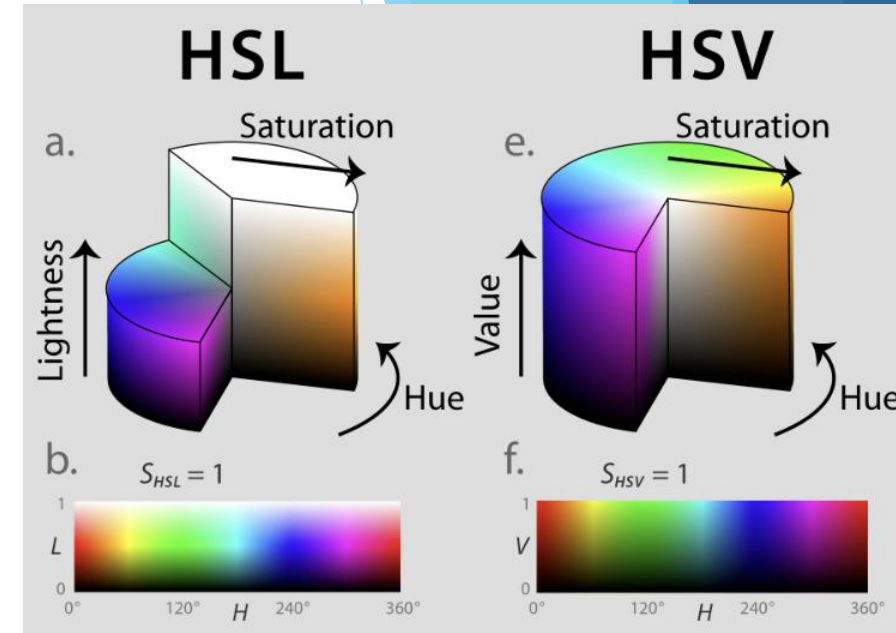
What color? Lightness or darkness value? level

Three dimensions of color

- ▶ A color sphere or color solid :
- ▶ Is the three dimensional representation of a color system where it depicts an added dimension of color variation. A 2D color wheel typically depicts the variables of hue and brightness, a color solid adds the variable of saturation, depicting all conceivable colors in an organized three dimensional structure.

Color representation

- CIE XYZ Color space
- CIE Chromaticity space
- HSL, HSV, LUV, CIE Lab



International Commission on Illumination (CIE)

Chromaticity is an objective specification of the quality of a color regardless of its luminance. Chromaticity consists of two independent parameters as hue (h) and colorfulness (s), where the latter is alternatively called saturation, chroma, intensity, or excitation purity

HSL (for hue, saturation, lightness)

HSV (for hue, saturation, value)

Lab It expresses color as three values: L^* for perceptual lightness and a^* and b^* for the four unique colors of human vision: red, green, blue and yellow.

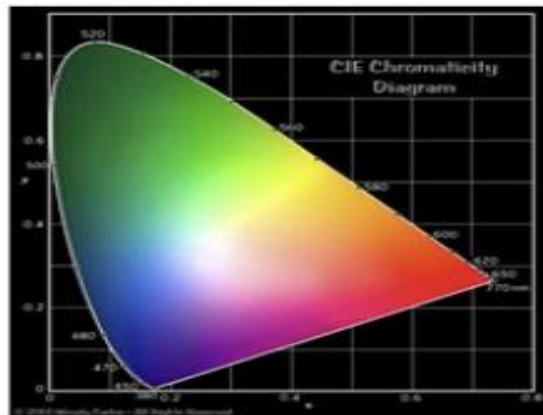
CIE XYZ Color Coordinate system

1931 – The **Commission International de l'Eclairage (CIE)**
Defined a standard system for color representation.

The CIE-XYZ Color Coordinate System.

In this system, the *XYZ Tristimulus values* can describe any visible color.

The XYZ system is based on the color matching experiments

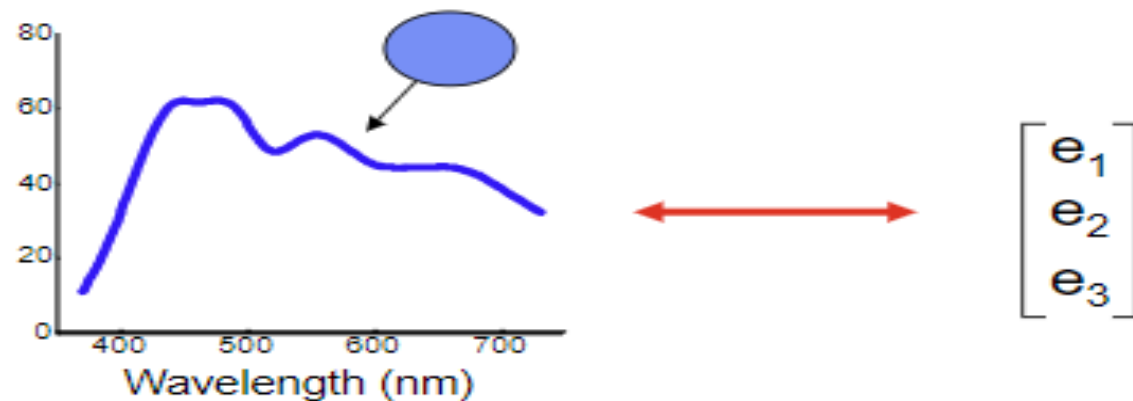


CIE XYZ Color Coordinate system

Trichromatic Color Theory

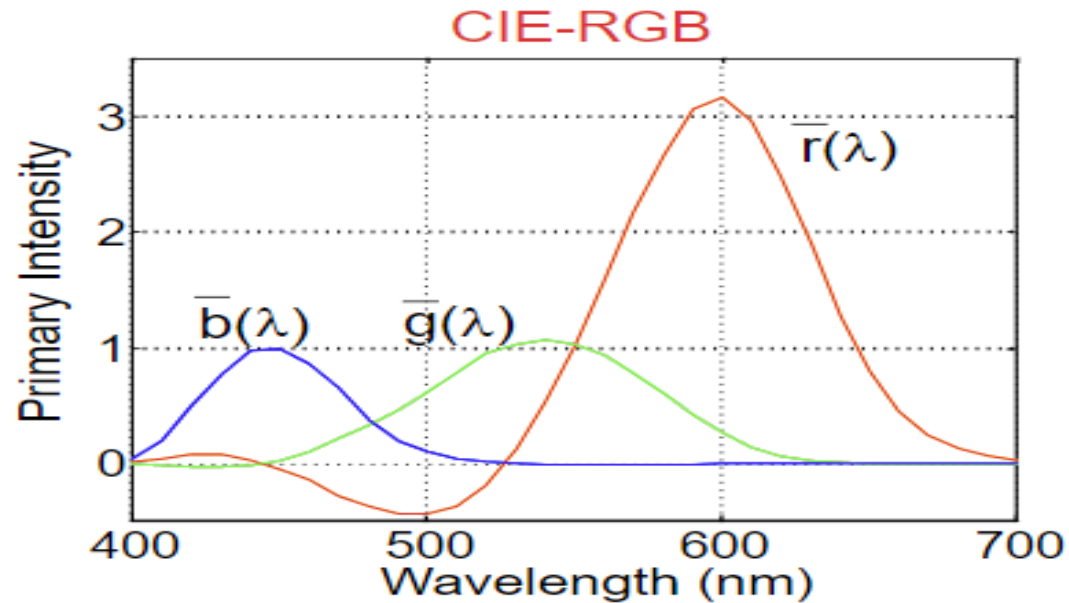
“tri”=three “chroma”=color

Every color can be represented by 3 values.



Space of visible colors is 3 Dimensional.

Calculate CIE XYZ Color Coordinate system



David Wright 1928-1929, 1929-1930 & John Guild 1931
17 observers responses to Monochromatic lights between 400-700nm using viewing field of 2 deg angular subtense.

Primaries are monochromatic : 435.8 546.1 700 nm
2 deg field.

These were defined as CIE-RGB primaries and CMF.

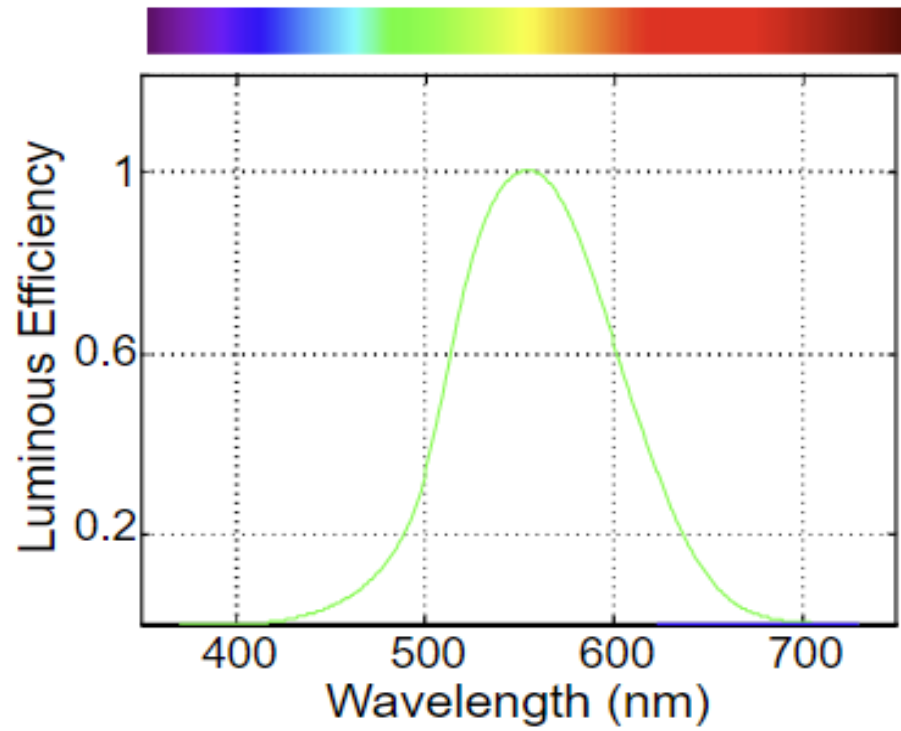
XYZ are a linear transformation away from the observed data.

Calculate CIE XYZ Color Coordinate system

CIE Criteria for choosing Primaries X, Y, Z and Color Matching Functions $\bar{X}, \bar{Y}, \bar{Z}$.

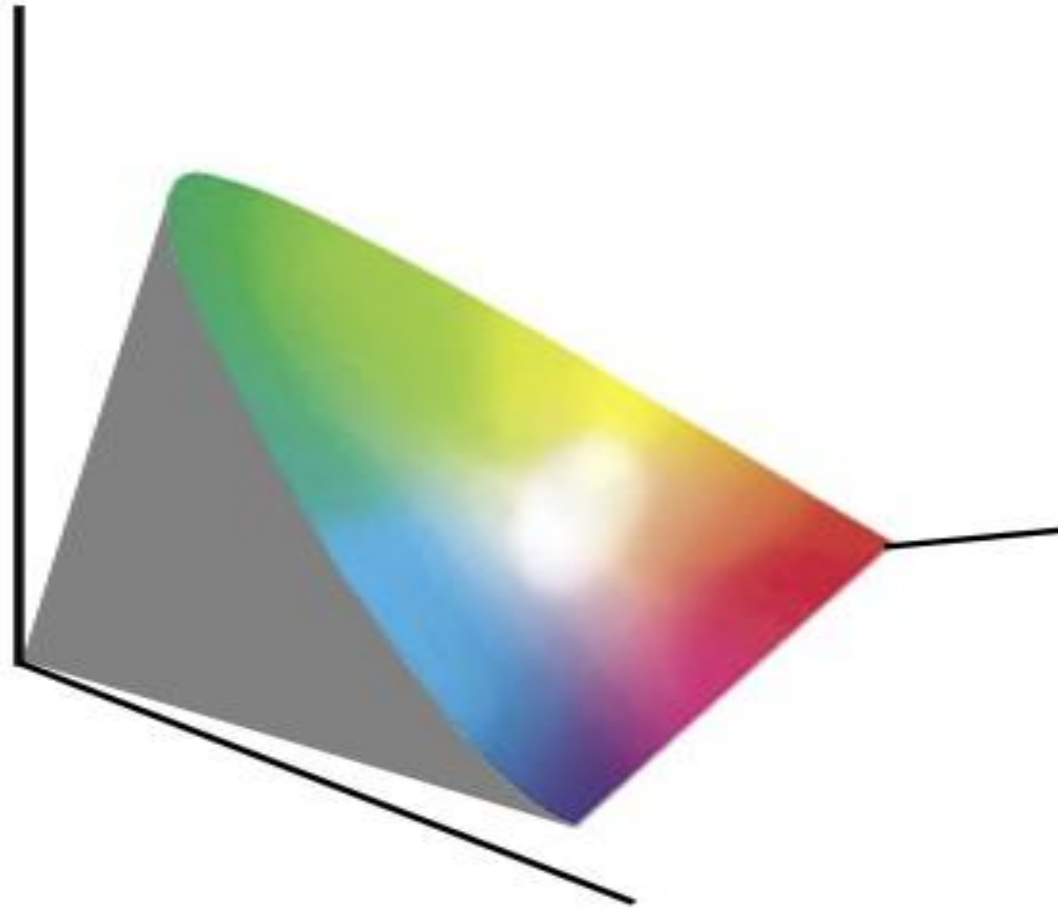
- 1) CMFs are non-negative over visible wavelengths.
(i.e. any color is represented by 3 positive values).
- 2) Equal amounts of the Primaries produce white.
(i.e. $X=Y=Z$ for stimulus of equal luminance at each wavelength).
- 3) The $\bar{y}$ color matching function is defined to match the luminous-efficiency function of the human eye.
- 4) Primaries are as 'tight' as possible around the set of possible colors (Maxwell triangle Projects to equilateral in XYZ space).

Luminous-Efficiency function of the human eye

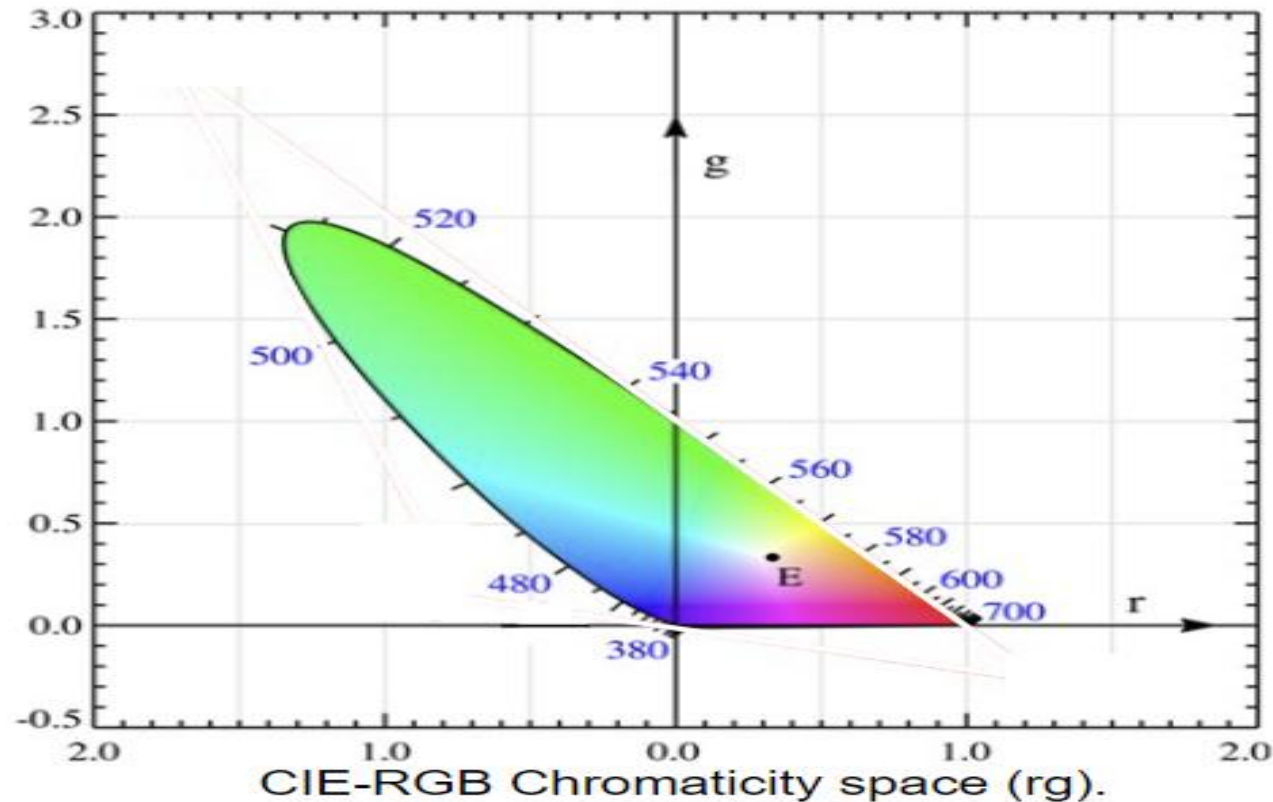


Color blindness changes the sensitivity of the eye as a function of wavelength. For people with protanopia, the peak of the eye's response is shifted toward the short-wave part of the spectrum (approximately 540 nm), while for people suffering deuteranopia, there is a slight shift in the peak of the spectrum, to about 560 nm. People with protanopia have essentially no sensitivity to light of wavelengths more than 670 nm.

CIE XYZ Color coordinate system



CIE-RGB to CIE-XYZ



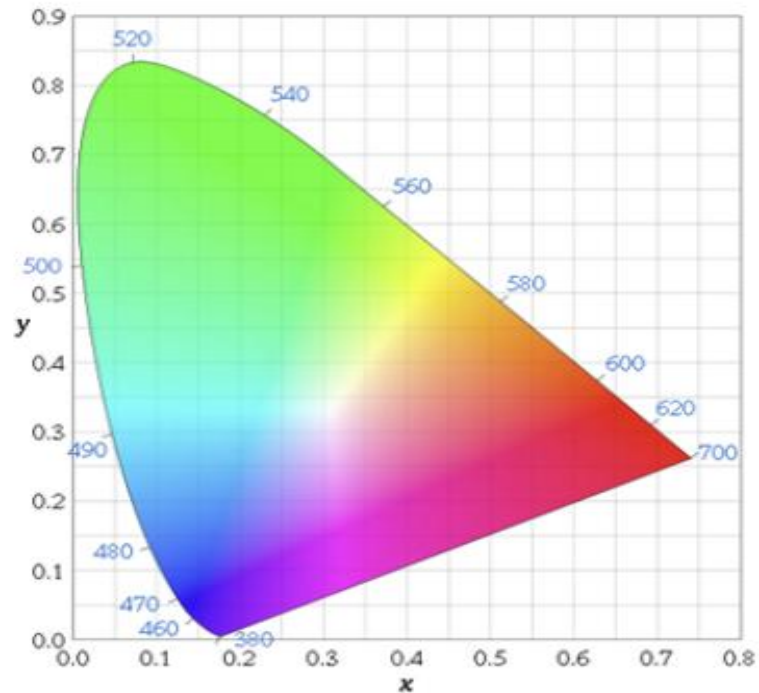
- * Cr, Cg, Cb must enclose the Gamut.
- * Line Cb-Cr is defined by Y being Luminance Function. (the Alychne = line of zero luminance).
- * Line Cr-Cg is tangent at 650+ (z is zero beyond 650).
- * Thus Cr is defined.
- * Equal Energy ($x=y=z=1/3$) puts constraint on Cb-Cg
- * Tight around Gamut $\rightarrow$ line Cb-Cg is close to green.
- * Cb and Cg are defined.

CIE-RGB to CIE-XYZ

CIE RGB space to XYZ space.

Map Cb Cg Cr to $x=(0,0)$ $y=(0,1)$ $z=(1,0)$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \frac{1}{b_{21}} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \frac{1}{0.17697} \begin{bmatrix} 0.49 & 0.31 & 0.20 \\ 0.17697 & 0.81240 & 0.01063 \\ 0.00 & 0.01 & 0.99 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



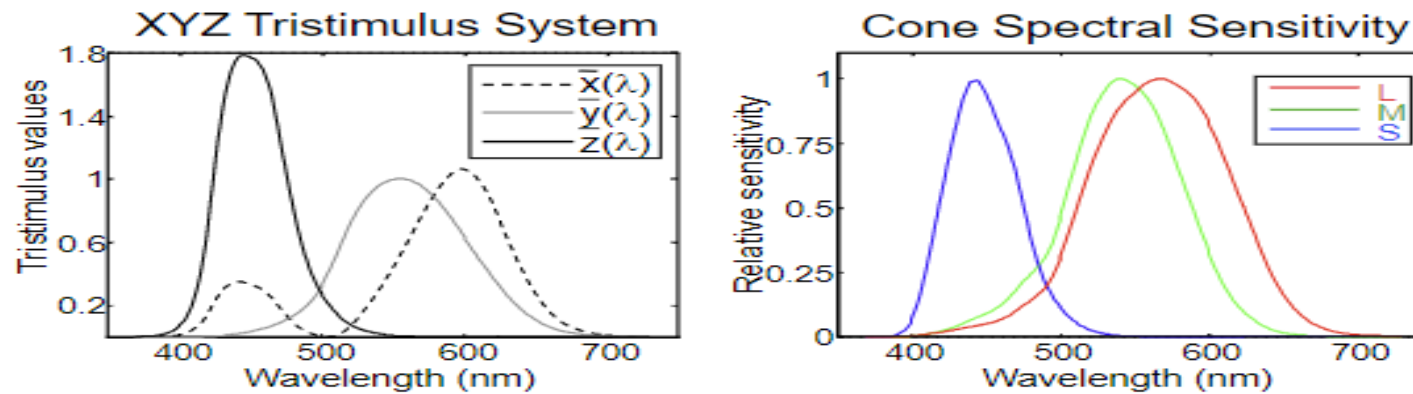
Colorimeters



A colorimeter is a light-sensing device that is used to measure the absorbance and transmittance of light as it moves through a sample of liquid. The main components of a colorimeter include: Illuminant: A specific, fixed light source that passes through the object.

analysis of blood, water, soil nutrients and foodstuffs, determination of solution concentration, determination of reaction levels, and determination of bacterial crop growth

Color matching function vs LMS(long medium short)-cone photoreceptor responses



The cone responses form a 3D linear system.
Cone responses are equivalent for metamers.

thus

The cone spectral sensitivities and the XYZ color matching functions are related by a 3 x 3 linear transformation.

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1.9023 & -1.4000 & 0.3544 \\ 0.6371 & 0.3933 & -0.0093 \\ 0.0007 & 0.0033 & 1.7462 \end{bmatrix} \begin{bmatrix} L \\ M \\ S \end{bmatrix}$$

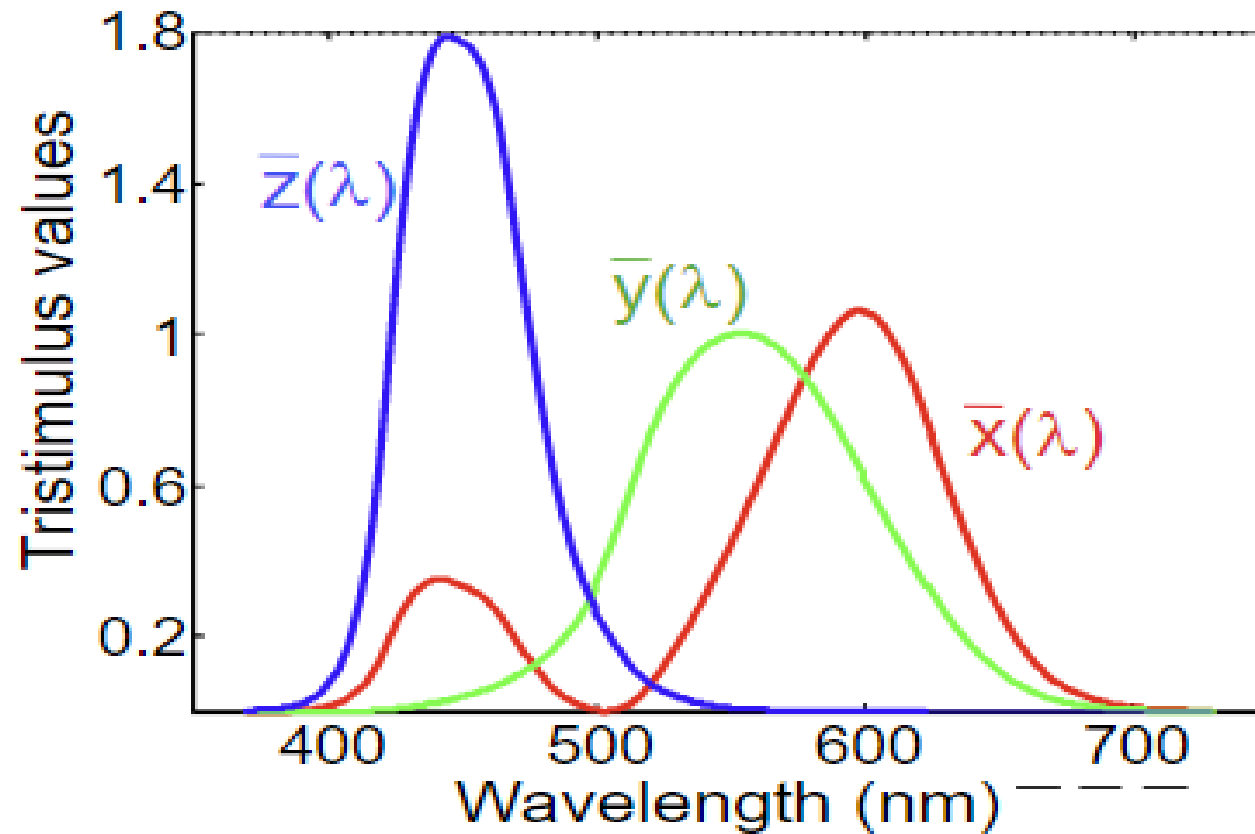
CIE -RGB

Primaries are monochromatic : 435.8 546.1 700 nm

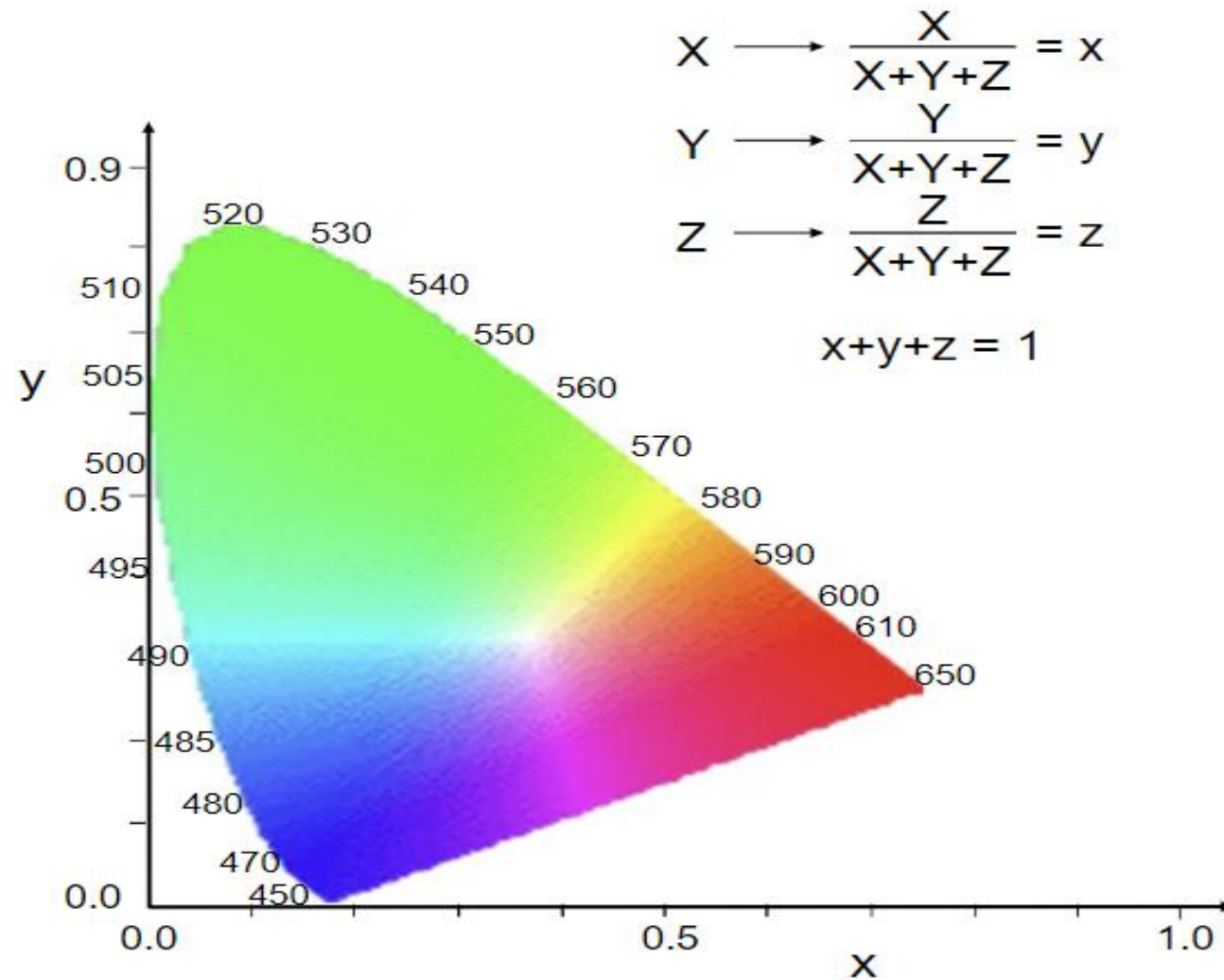
$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.49 & 0.31 & 0.20 \\ 0.17 & 0.81 & 0.01 \\ 0.00 & 0.01 & 0.99 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

CIEXYZ Color Coordinate System

$\bar{x}$ $\bar{y}$ $\bar{z}$ Color Matching Functions

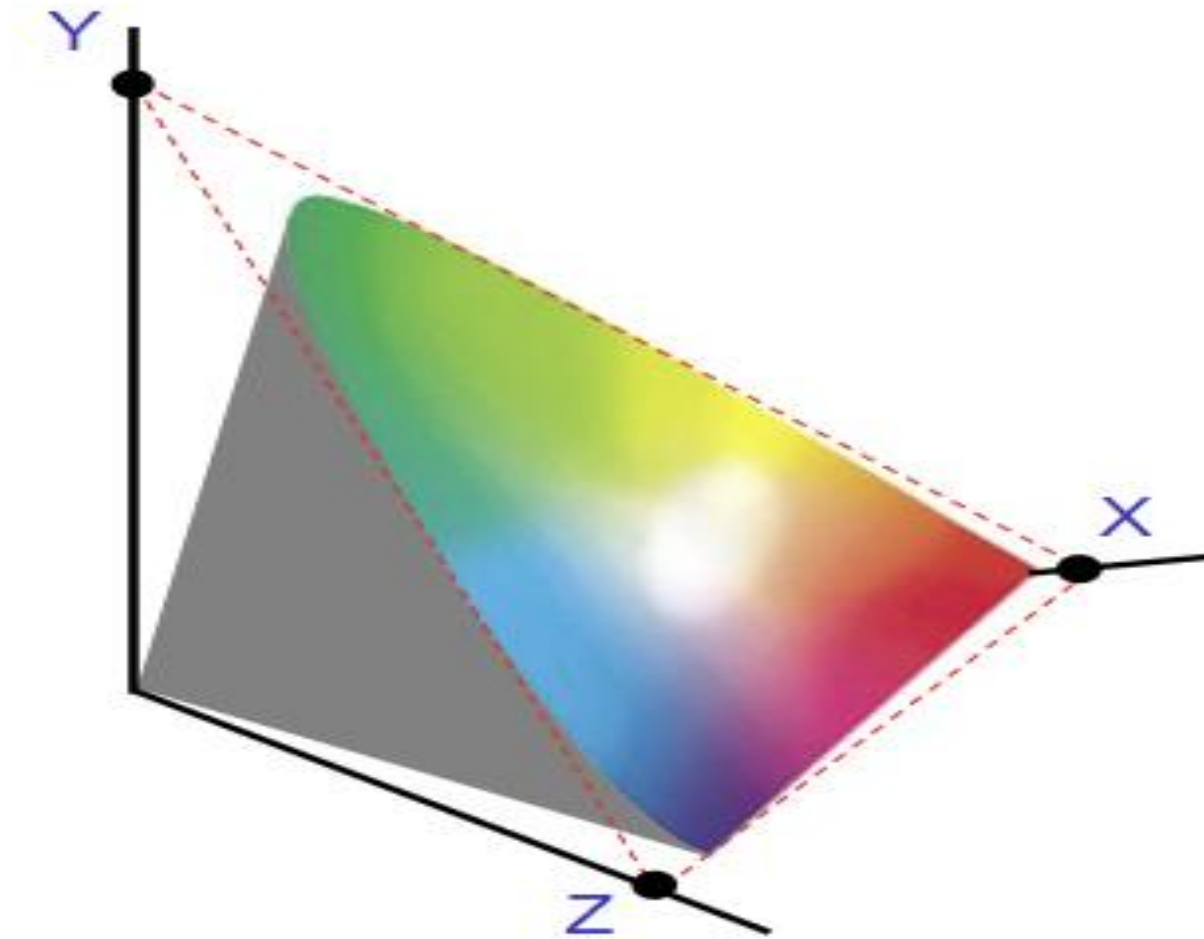


CIE Chromaticity Diagram

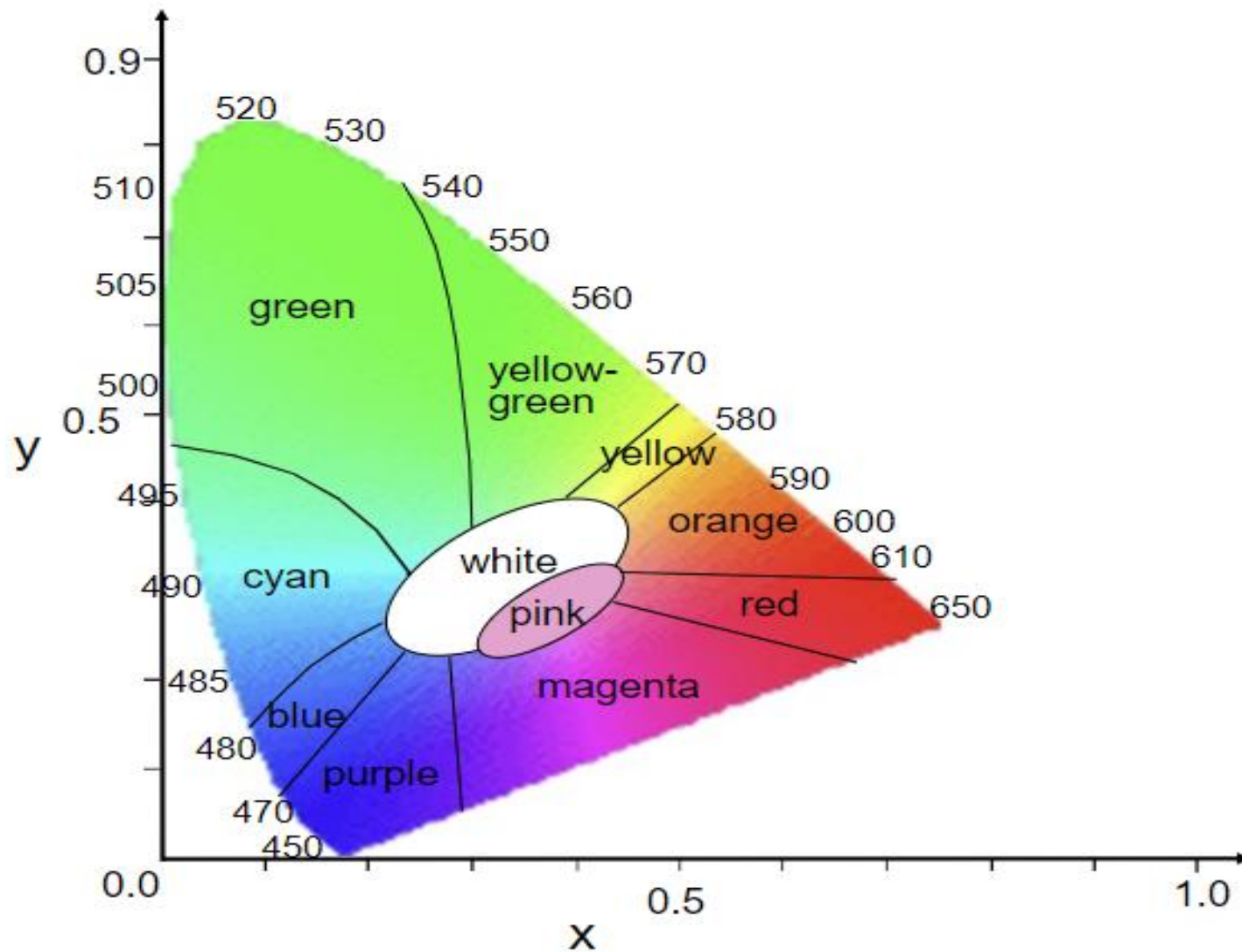


A common representation of color signal

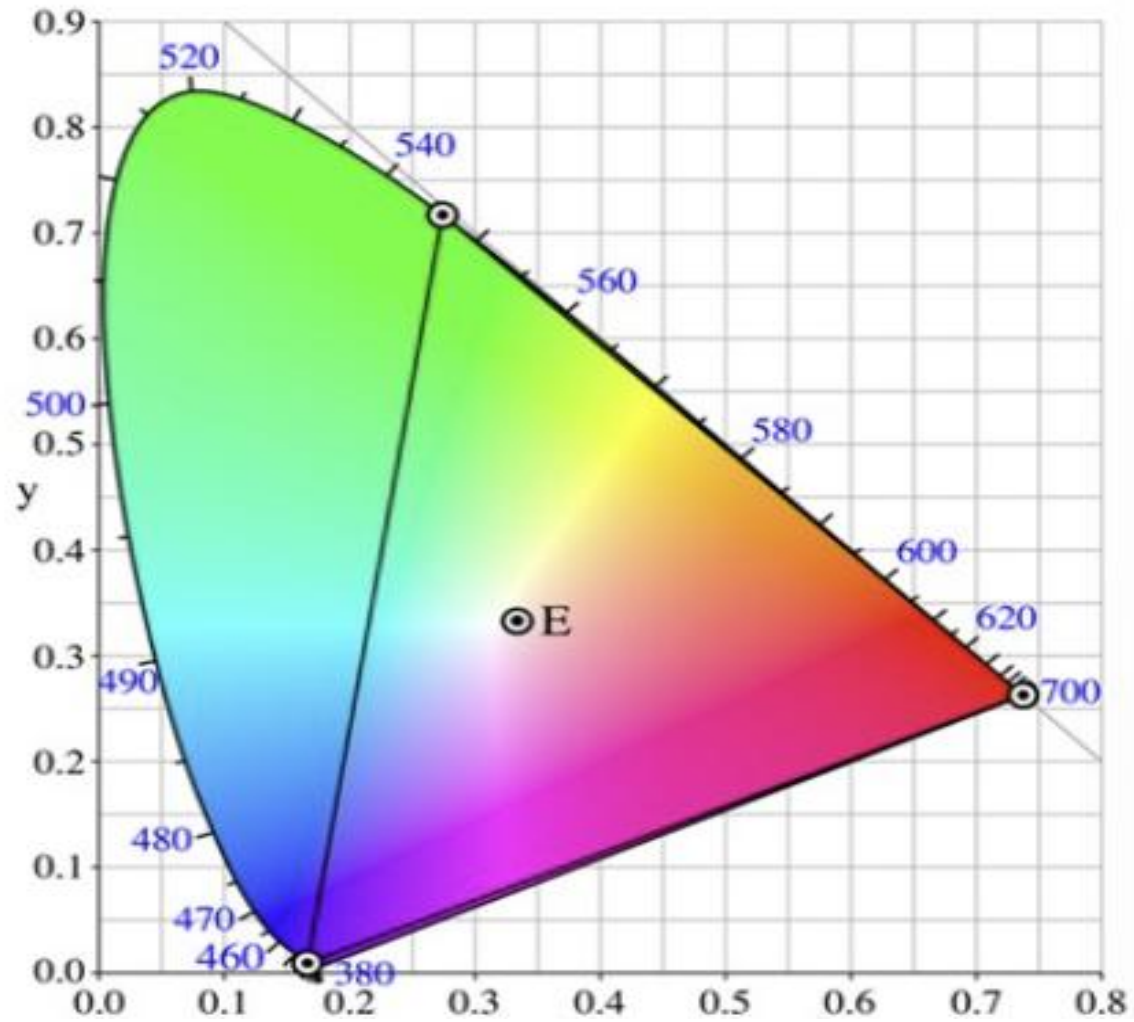
CIE Chromaticity Diagram



Color Naming



CIE-RGB Primaries



Blackbody Radiators and CIE standard Illuminants

CIE Standard Illuminants:

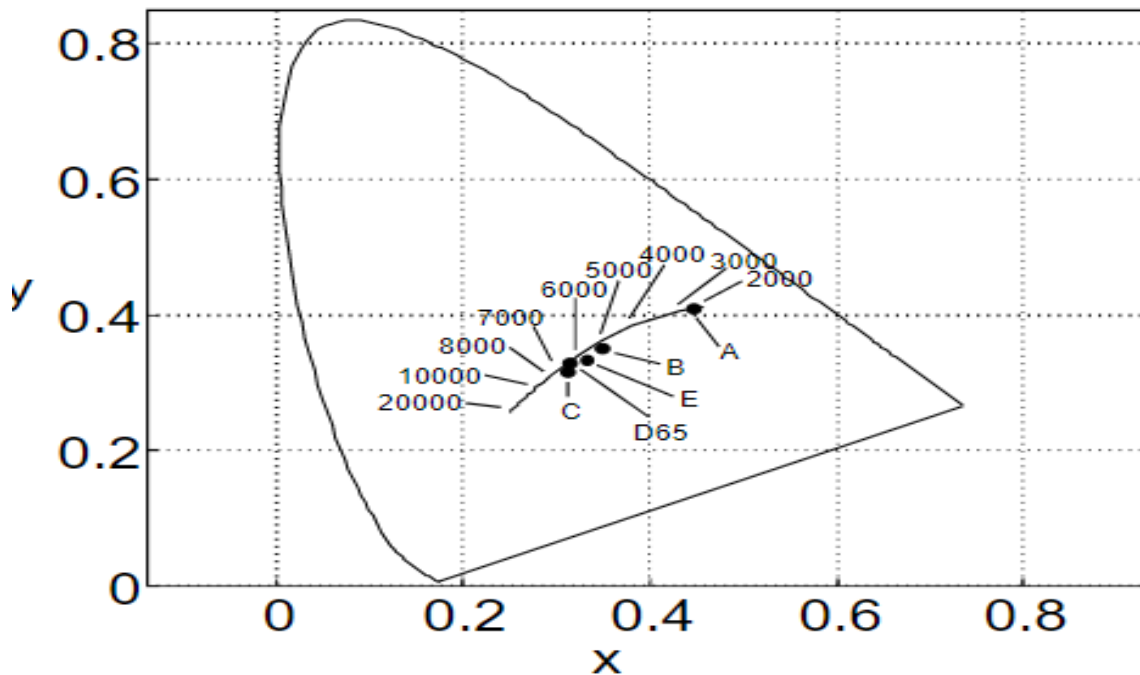
A - tungsten light

B - Sunset

C - blue sky

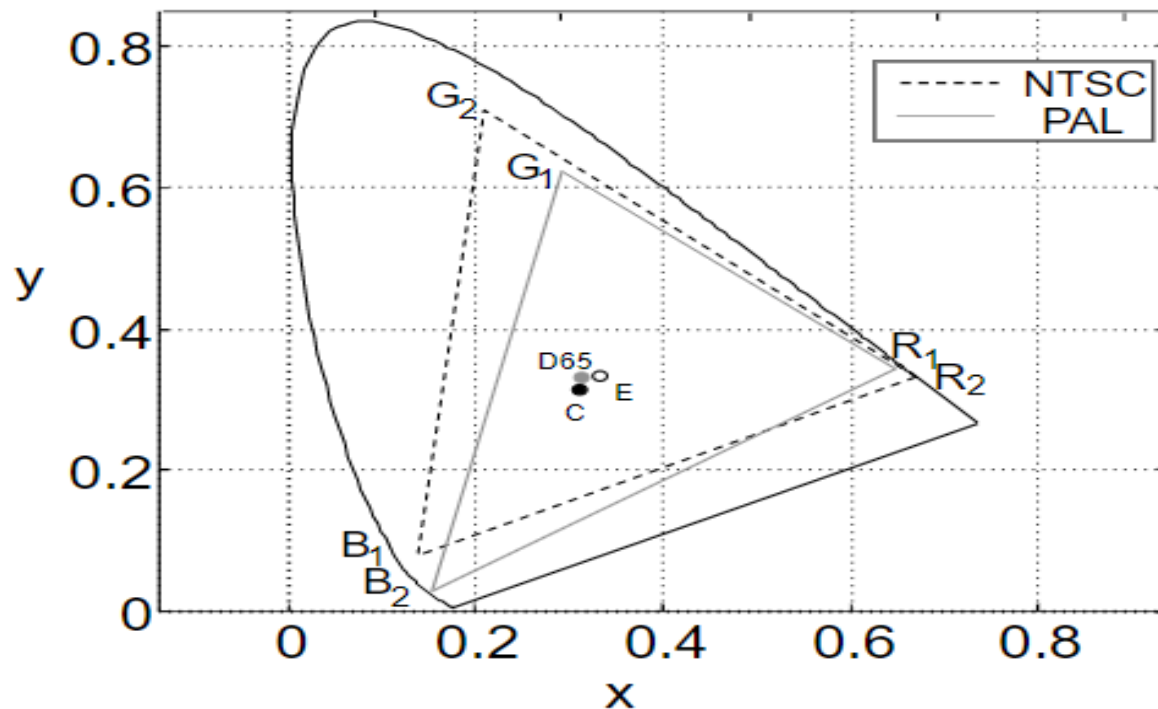
D65 - Average daylight

E - Equal energy white ($x=y=z=1/3$)



Television Primaries and Gamut

$R_1G_1B_1$ - Primaries used for PAL
 $R_2G_2B_2$ - Primaries used for NTSC
D65 - reference white for PAL
C - reference white for NTSC



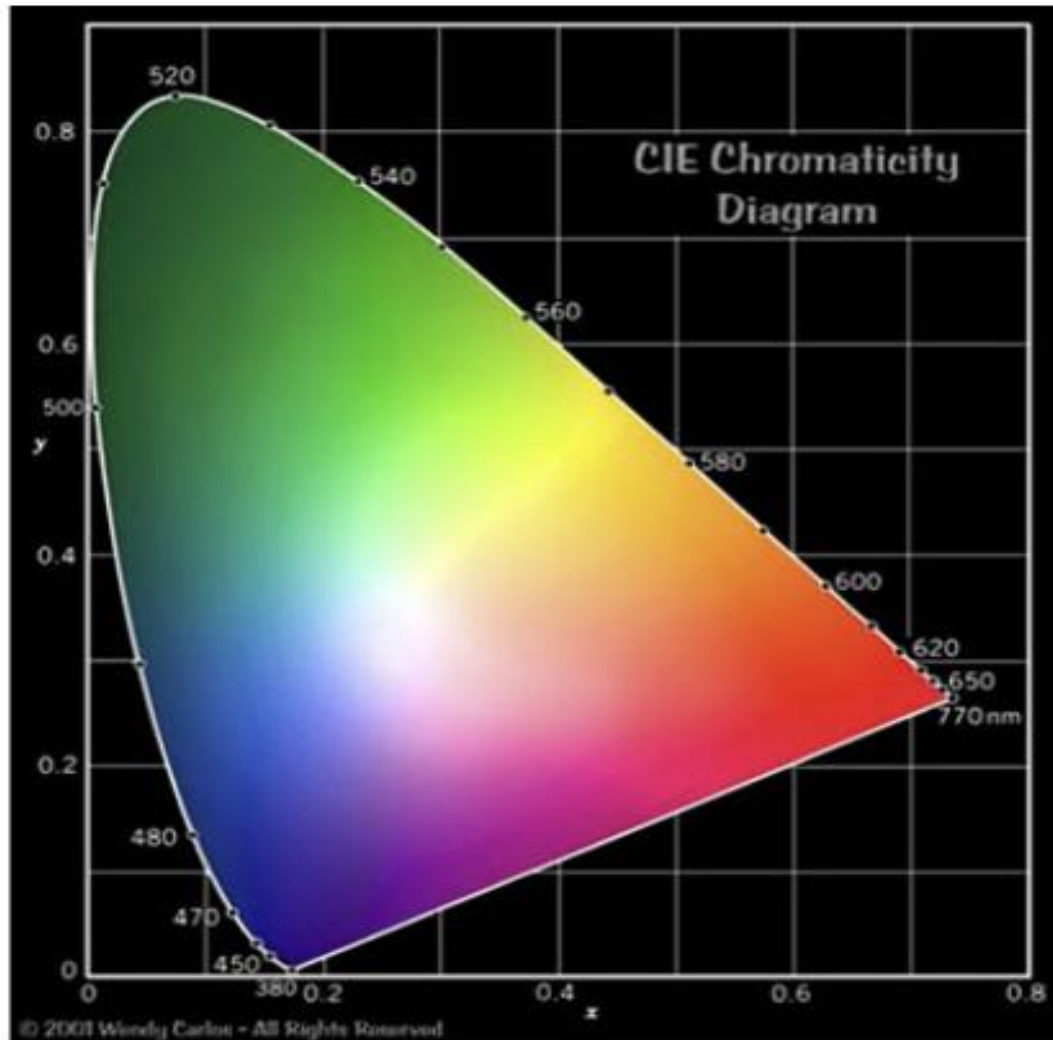
Phase Alternating Line (PAL)

National Television System Committee

Gamut is a certain complete subset of colors. The most common usage refers to the subset of colors which can be accurately represented in a given circumstance, such as within a given color space or by a certain output device

NTSC televisions broadcast 525 lines of resolution, while PAL televisions broadcast 625 lines of resolution.

XYZ Color space Hue vs Saturation

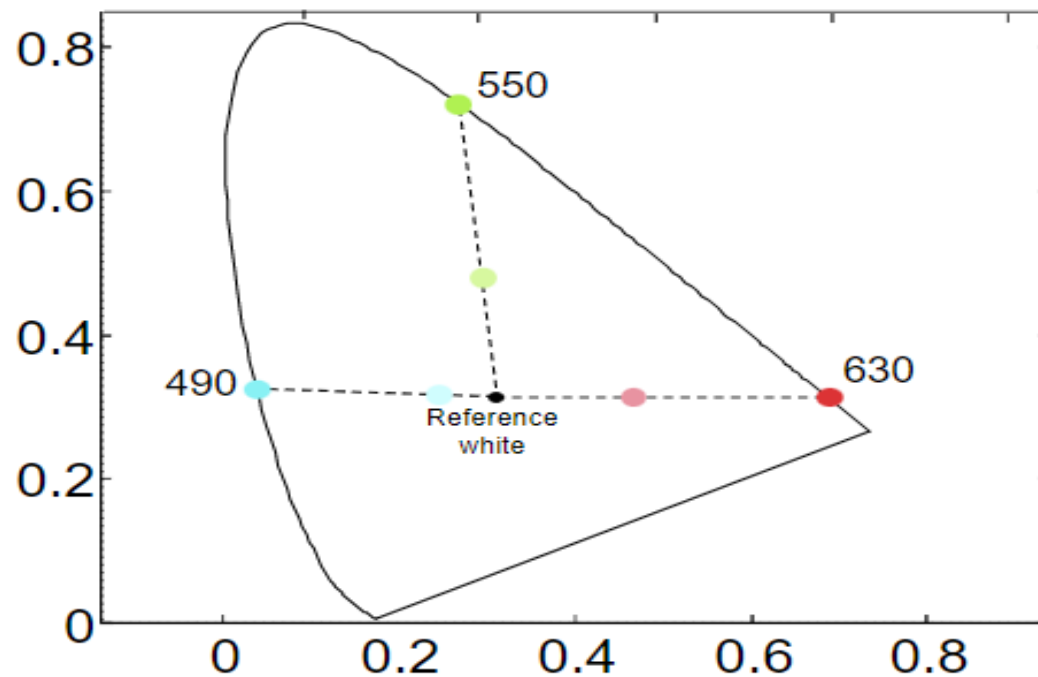


Chromaticity in Polar Coordinates

Given a reference white.

Dominant Wavelength —

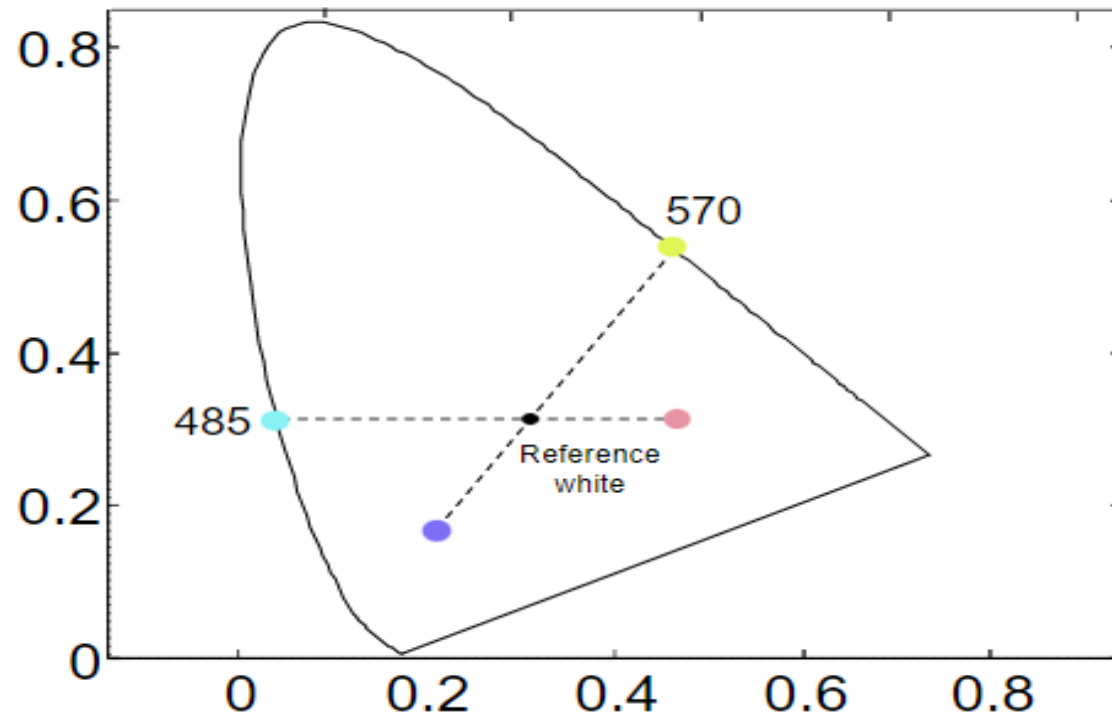
wavelength of the spectral color which added to the reference white, produces the given color.



Chromaticity in Polar Coordinates

Given a reference white.

Complementary Wavelength —
wavelength of the spectral color which added
to the given color, produces the reference white.

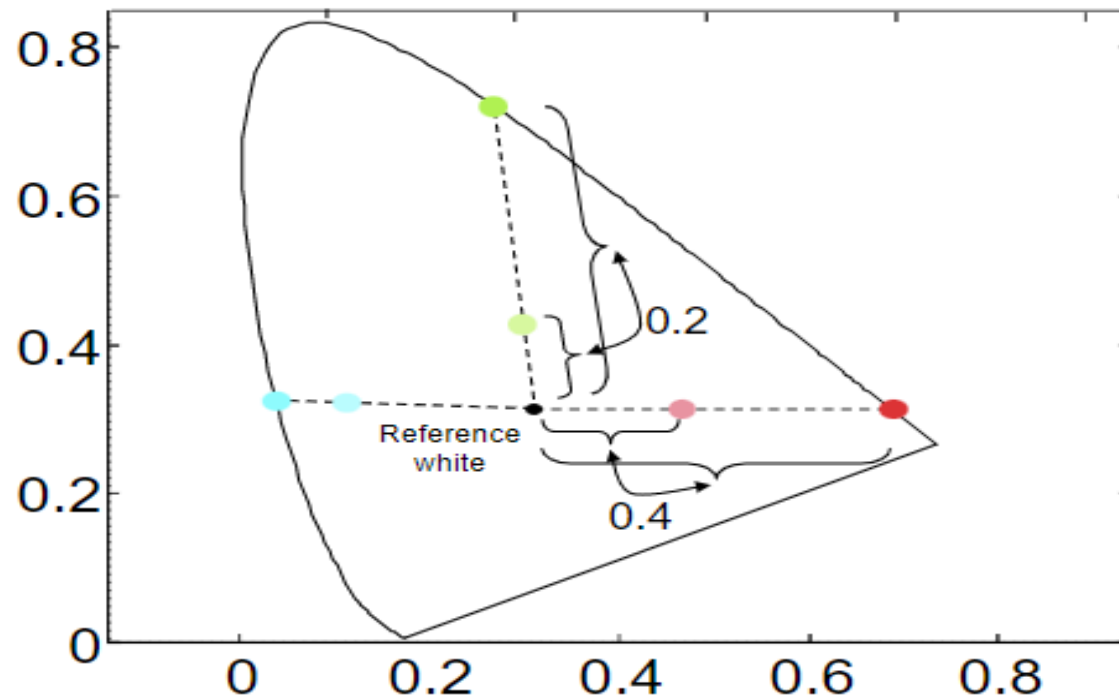


Chromaticity in Polar Coordinates

Given a reference white.

Purity –

the ratio of the lengths between the given color and reference white and between the dominant wavelength and reference white. Ranges between 0 .. 1.



Chromaticity in Polar Coordinates

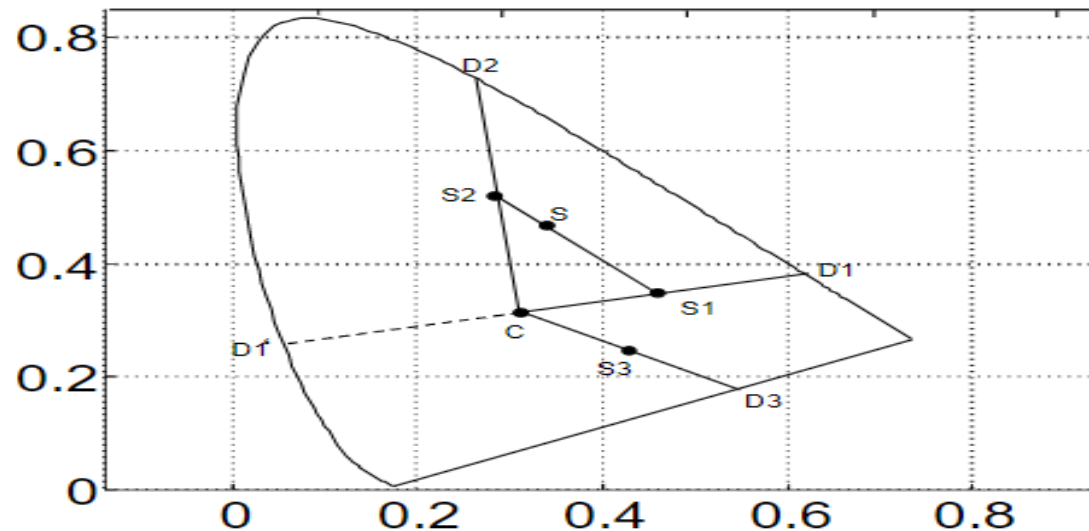
EXAMPLE:

Reference white is CIE standard illuminant - C.

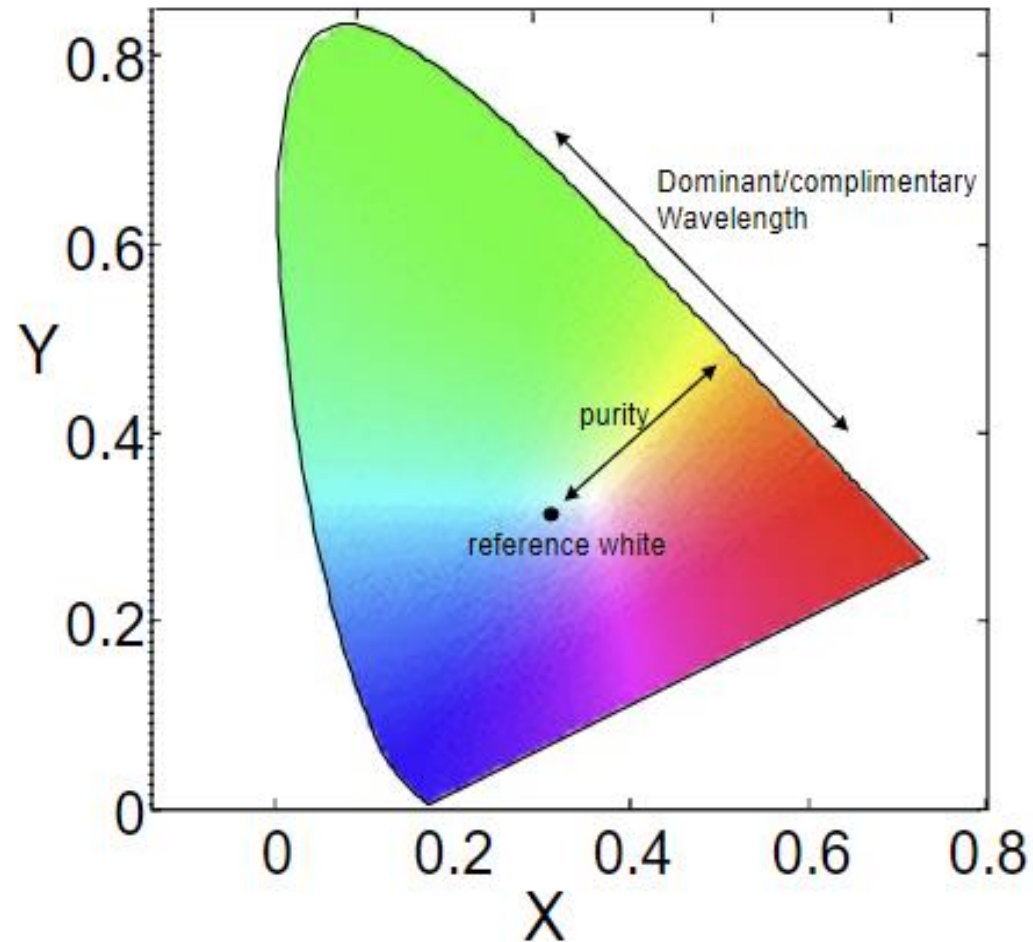
Dominant Wavelength of color S_1 is D_1
of color S_2 is D_2 .

Complementary Wavelength of color S_1 is D_1 .
 S_2 does not have a complimentary wavelength.

Excitation Purity of S_1 is the ratio CS_1/CD_1
of S_2 is the ratio CS_2/CD_2
of S_3 is the ratio CS_3/CD_3



Chromaticity in Polar Coordinates



Chromaticity is an objective specification of the quality of a color regardless of its luminance. Chromaticity consists of **two independent parameters**, often specified as **hue (h)** and **colorfulness (s)**, where the latter is alternatively called saturation, chroma, intensity, or excitation purity.

Color description

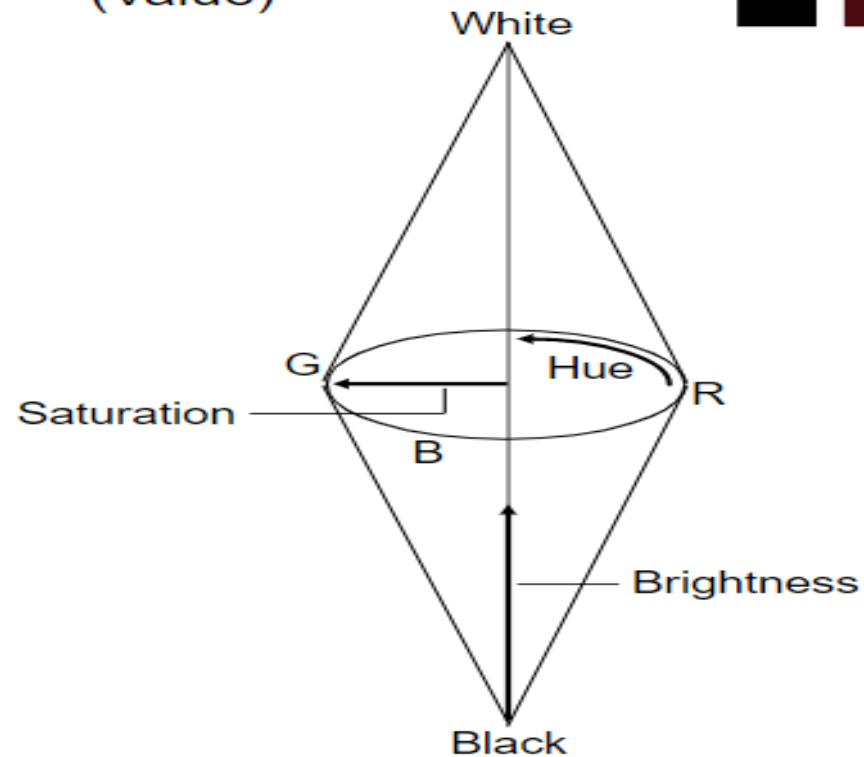
Hue (red, green, yellow, blue ...)



Saturation (pink, bright red,)



Lightness (black, grey, white)
(Value)



Color stimuli

- ▶ Color stimuli can be of different types as shown in [Figure 9.4](#). When it has light of only one wavelength, it is called *monochromatic*, e.g. a laser beam. When the relative amount of light from all wavelengths is equal, then it is called *achromatic*. The sunlight is close to achromatic in the day time. Finally, if the stimulus has different amounts of light from different wavelengths, it is called *polychromatic*. Most of the time we deal with polychromatic light. Most of the man made and natural color stimuli are smooth spectra of polychromatic light.

Color stimuli

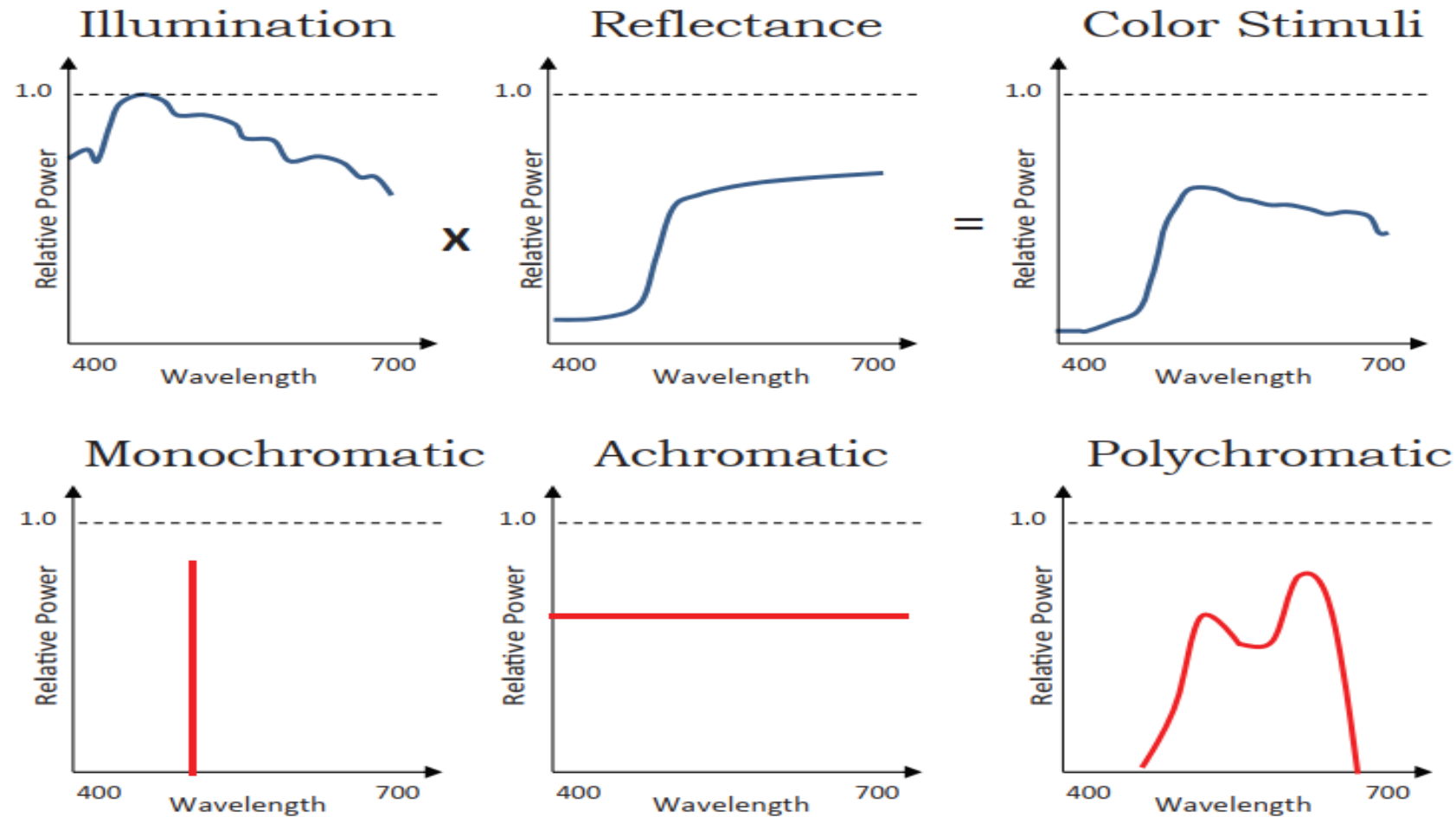


Figure 9.4. Top: The product of the illumination and the reflectance spectrum generate the color stimuli. Bottom: This shows different types of color stimuli - from left to right - monochromatic, achromatic and polychromatic.

Reference

► Textbook

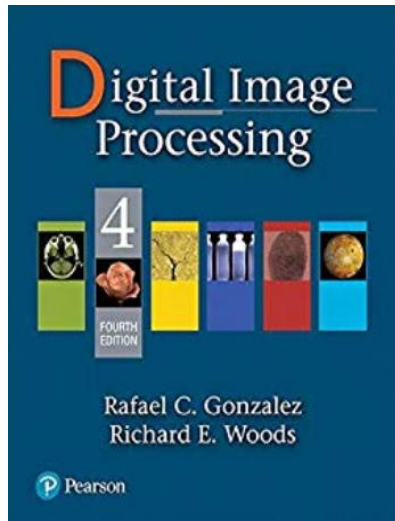


Image processing

Question



Thank you

